		Mid Sem Spring 2025 Question Papers DPS			
	Course Code	Course Name	Type	Credits	Instructor(s)
	PH1201	Electricity and Magnetism	Theory	3	Ananda Dasgupta,
					Siddhartha Lal
	PH2201	Basic Quantum Mechanics	Theory	3	Amit Ghosal
	PH2202	Thermal Physics	Theory	2	Koushik Dutta
	PH2204	Mathematical Methods of Physics I	Theory	2	Pradeep Kumar Mohanty
	PH3201	Basic Statistical Mechanics	Theory	4	Anandamohan Ghosh
	PH3202	Intermediate Electricity and Magnetism	Theory	4	Rajesh Kumble Nayak
	~~PH3203~~	~~Advanced Quantum Mechanics~~	Theory	4	Sourin Das
	PH4202	Advanced Statistical Mechanics	Theory	4	Bhavtosh Bansal
	PH4203	Research Methodology	Theory	4	Soumitro Banerjee
	PH4204	High Energy Physics	Theory	4	Ritesh Kumar Singh
	PH4205	General Theory of Relativity and Cosmology	Theory	4	Golam Mortuza Hossain
	PH4207	Quantum Information Processing	Theory	4	Chiranjib Mitra
	PH4209	Evolutionary Dynamics	Theory	4	Supratim Sengupta
	PH4213	Symmetry Methods in Physics	Theory	4	Ananda Dasgupta

Siddhartha Lal
1/3/25

PH1201 Electricity & Magnetism **Mid-semester Examination**

Instructor: Dr. Siddhartha Lal **Spring Semester, 2025**

Instructions: Answer all questions for full marks. Put your name and roll number on both Question paper and Answer script, and hand both in. $e = -1.6\times10^{-19}$C, $1/4\pi\epsilon_0 = 9\times10^9 Nm^2/C^2$

Date: March 1, 2025 **Full marks: 20** **Time: 1.5 hours**

Name: **Roll Number:**

Q1. Computing Electric Flux. [6 marks]

Consider a cylinder of radius R and height L with it's center at the origin of coordinates and axis aligned along $\hat{k}$. For an electric field $\vec{E} = \frac{E_0}{\sqrt{2}}(\hat{i}+\hat{j}+\hat{k})$ where E_0 is a constant, compute the electric flux through
(i) the upper flat surface [2 marks],
(i) the lower flat surface [1 mark], and
(ii) the curved surface of the cylinder [3 marks].

Q2. Electric field of a disk. [10 marks]

(i) An infinitesimally thin disk of radius R contains a uniform surface charge density σ, and is placed in the XY plane with it's center at the origin of coordinates. Find the electric field at an axial point lying at a distance z above the center of the disk. Compare your answer with the electric field due to an infinite plane of uniform surface charge density σ (you do not have to derive this!). [5 marks]
(ii) An infinitesimally thin disk of radius R contains a surface charge density that varies linearly with the radial distance on the disk $\sigma = \sigma_0 r/R$ (and σ_0 is a constant), and is placed in the XY plane with it's center at the origin of coordinates. Find the electric field at an axial point lying at a distance $z >> R$ above the center of the disk. [5 marks]

Q3. Playing with charges. [4 marks]

Consider two point charges q and λq (where λ is a dimensionless parameter) located at the points $x = a$ and $x = \mu a$ respectively. Assuming that the sum of the two charges is a constant, compute the value of λ for which the magnitude of the electrostatic force between the two charges is maximum.

Siddhartha Lal
(on behalf of
Amit Ghosal)
25/2/25

Mid-semester Exam, PH2201 (25 Feb 2025)

(1) Let a normalized wave function Ψ for a particle be given by,

$$\Psi(x) = A \exp\left(-|x|/L\right)$$

(a) Sketch $\Psi(x)$ and find A. (b) Calculate the expectation values $\langle \hat{x} \rangle$, $\langle \hat{x}^2 \rangle$ in the given Ψ. (c) What is the probability to find the particle in the region $(0, L)$ and $(L, +\infty)$? Here, $(0, L)$ means "x values between 0 and L".
Marks: 1+2+2=5

(2) The normalized eigen-functions and eigen-energies of a particle in an one dimensional infinite square well potential, defined by:

$$V(x) = 0 \quad \text{for } 0 \leq x \leq a, \quad \text{and } V(x) = \infty \text{ otherwise; are given by :}$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad \text{and} \quad E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}, \quad \text{where } n = 1, 2, 3 \cdots$$

Suppose a particle in such an infinite square well potential is described by the following normalized wave-function (at a particular time, say, at $t = 0$):

$$\Psi(x) = A\left[2\sin\left(\frac{\pi x}{a}\right) + \sin\left(\frac{2\pi x}{a}\right)\right]$$

where, A is an appropriate normalization constant.
(a) If the energy of the particle is measured by an observer, what will be the result? (b) If the energy of the particle is measured again (second time, in quick succession), what will be the result? (c) Instead of the measurement defined in (a), if a very large number of independent observers are making the energy measurement of the particle described by the above $\Psi(x)$, what will be the *average* energy recorded? Please justify.
(d) Suppose that a different particle in the same infinite square well is described by $\Psi(x) = C$, where C is a non-zero constant for $x \leq a/2$, and $\Psi(x) = 0$ for $a > x > a/2$. Sketch $\Psi(x)$. If the energy of this particle is now measured, what is the probability of finding it in the ground state?
Marks: 1+0.5+2+3=6.5
[Additional information (not needed for producing your solution): If you are worried that the given $\Psi(x)$ in part (d) is not continuous within its domain, please note that the continuity is demanded for eigenfunctions, not wave functions. Also, this $\Psi(x)$ can be expanded (in a limiting sense) as a linear combination of eigenfunctions (remember the completeness of eigenfunctions discussed in class.]

(3) Consider simple harmonic oscillator defined by,

$$\hat{\mathcal{H}} = \hbar\omega\left(\hat{a}_\pm \hat{a}_\mp \pm \frac{1}{2}\right), \quad \text{where,} \quad \hat{a}_\pm = \frac{1}{\sqrt{2\hbar m\omega}}\left(\mp i\hat{p} + m\omega\hat{x}\right).$$

It is the same expressions used in the class. Here, $\hat{\mathcal{H}}\psi_n = \hbar\omega\left(n + \frac{1}{2}\right)\psi_n$, where ψ_n are the eigenfunctions of $\hat{\mathcal{H}}$with corresponding eigenenergies. Also given: $\hat{a}_+\psi_n = c_n\psi_{n+1}$, $\hat{a}_-\psi_n = d_n\psi_{n-1}$, here c_n and $d_n$are n-dependent constants. Further, assume the identity,

$$\int_{-\infty}^{+\infty} \left(\hat{a}_\pm\psi_n\right)^* \left(\hat{a}_\pm\psi_n\right) dx = \int_{-\infty}^{+\infty} \left(\hat{a}_\mp\hat{a}_\pm\psi_n\right)^* \psi_n dx.$$

(a) Derive the constants c_n and d_n explicitly as a function of n.
(b) Using all given information, evaluate the expectation value $\langle \hat{x}^4 \rangle$ in the nth eigenfunction.
Marks: 2+6.5=8.5

PH2202 Thermal Physics
Fall Semester - 2025
Indian Institute of Science Education and Research, Kolkata
Instructor: Koushik Dutta

Mid-semester Examination Date: 03/03/2025

No clarification regarding the questions will be provided. If any genuine issue is found in any problem, marks will be given appropriately to all.

1. (i) Draw a Carnot engine in the p -V diagram and show clearly adiabats, isotherms and heat flow. Mark each corner of the diagram as $A, B, C...$ **Marks: 2**

   (ii) Draw the same process in the T (y-axis)-S (x-axis) diagram and identify the corresponding corners $A, B, C..$ of part (i). **Marks: 2**

   (iii) Find out the entropy change in each part of the diagram in part (ii) and use that to calculate the efficiency of the engine. **Marks: 4**

2. In some calculations, you found that the entropy of a thermodynamic system is $S = B^{1/3}(NU/V)^{2/3}$, where B is a constant, N, V, U are the total number of molecules, volume and the internal energy of the system, respectively. Your friend made a cursory look at the final expression and claimed that the expression must be wrong. What might be the basis of her/his claim? **Marks: 3**

3. Consider a thermally isolated system of a fixed volume V, containing some gas. The volume V is divided in two compartments by a movable wall that allows exchange of energy between the two compartments. Initially, pressures and temperatures of the two compartments are unequal, and we allow the system to evolve. Find out the the equilibrium configuration of the system by minimising or maximising relevant thermodynamic quantity. **Marks: 7**

4. (i) For a general hydrodynamic system described by (p, V, T) show the following expression by demanding that entropy is a state function: **Marks: 5**

$$\frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_T + \frac{p}{T} = \left(\frac{\partial p}{\partial T}\right)_V .$$

   (ii) Find out the most general expression for the internal energy U for an Van der Wall's gas described by $p = \frac{nRT}{V-nb} - \frac{n^2 a}{V^2}$. **Marks: 2**

5. (i) Starting from the defination of the Gibbs Free energy $G = H - TS$, find out one of the four Maxwell's relations. **Marks: 3**

   (ii) Using the above relation, show that for a given T, C_p is independent of p for an ideal gas. **Marks: 2**

6. Consider the following expression for entropy of a thermodynamics system in equilibrium: $S = A^{1/3}(NVU)^{1/3}$. Here, A is a constant and other letters have their usual meaning in thermodynamics with N being the total number of molecules that remains fixed. Find out the equation of state for the system, *i.e.*, find out the function $f(p, V, N, T) = 0$. **Marks: 5**

7. The fundamental equation of a thermodynamic system is given by $U = AV^{-2}e^{S/R}$. The initial temperature and pressure are T_0 and p_0 respectively. The system undergoes an isentropic change ($dS = 0$) so that the pressure is halved. Find the final temperature. **Marks: 5**

Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)
Midsem Exam, Date: 7 March, 2025

IISER KOLKATA

Timing: 15:00 to 16:30 | Spring 2025 | Max mark: 100

Solve all questions, each carry [20] marks.

Notation: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \equiv (x, y, z) \equiv (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$; $\nabla \equiv (\partial_x, \partial_y, \partial_z) \equiv (\partial_r, \frac{1}{r}\partial_\theta, \frac{1}{r\sin\theta}\partial_\phi)$.

$A = |\vec{A}| = \sqrt{\vec{A}.\vec{A}}$ $\quad x \equiv x(t)$, $\dot{x} = \frac{d}{dt}x(t) \equiv Dx(t)$, $\ddot{x} = \frac{d^2}{dt^2}x(t) \equiv D^2x(t)$, $i = \sqrt{-1}$

1. (a) Express ∇ in (r, θ, ϕ)-coordinate system, where $x = r\cos\theta\sec\phi, y = r\sin\theta\sec\phi, z = r\tan\phi$. **[10]**
   (b) Use Levi Civita symbols to prove $\vec{A} \times (\nabla \times \vec{A}) = \frac{1}{2}\nabla A^2 - (\vec{A}.\nabla)\vec{A}$ **[10]**

2. (a) Find an analytic function $f(z)$ whose real part is $x^2 + xy - y^2$. **[10]**
   (b) Find regions in complex plane where both $f_1(z) = |y| - ix$ and $f_2(z) = |x| + i|y|$ are analytic. **[10]**

3. Find the unit vectors which are parallel to the tangent line of the curves $\Re(ze^z)$ and $\Im(ze^z)$ at $z = z_0 = 1 + i$, and show that they are orthogonal. Use Cauchy–Riemann criteria to show that $z_0 = 1 + i$ is not special, the orthogonality holds at any other arbitrary point $z_0 = a + ib$. **[20]**

4. (a) Use variable transformation to convert the equations $\{\dot{x} = x\ln y; \dot{y} = -y\ln x\}$ to linear form and solve them using matrix method for initial condition $x(0) = e^2$ and $y(0) = 1/e$ **[10]**
   (b) Show that particular integral of the diff eqn $(D - a)x(t) = f(t)$ is $e^{at}\int_0^t f(t')e^{-at'}dt'$. **[10]**

5. (a) Given a matrix $M = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, compute M^n **[12]** (b) What is $\lim_{n\to\infty} M^n$? **[8]**

PH3201 Mid Semester Examination

21 February 2025 10:00-11:30AM

Attempt all questions.

1. Find the leading term of the integral for

$$\lambda \gg 1: \qquad I(\lambda) = \int_{-5}^{+4} \exp(-\lambda\, x^2)\, \log(1 + x^2)\, dx$$ [5]

2. Suppose there are two independent integer random variables $\{k_1, k_2\}$, each drawn from a probability distribution function $p(k) = \dfrac{\lambda^k \exp(-\lambda)}{k!}$ with parameters λ_1 and λ_2 respectively.

Determine the characteristic function of the random variable $(k_1 + k_2)$.

Hence find mean of $(k_1 + k_2)$ and variance of $(k_1 + k_2)$. [5]

3. Let the internal energy of a particular system be given by $U = A\dfrac{N^3}{V^2} \exp\left(\dfrac{S}{N\, k_B}\right)$ where $A(joule\ meter^2)$ is a constant. Initially the system is at temperature $T = 320\ K$ and at pressure $P = 2 \times 10^5\ Pascal$. The system expands reversibly, such that the final pressure is $P = 10^5\ Pascal$, by a process in which entropy does not change.

Find the final temperature. [4]

4. Consider a collection of N hard spheres confined in a box of volume V. Each sphere in a box occupies a tiny volume ϵ and apart from hardcore exclusion there are no other interactions.

Calculate the heat capacity ratio: $\gamma = \dfrac{C_P}{C_V}$ [6]

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \qquad C_P = T\left[\frac{\partial S}{\partial T}\right]_{P,N}, \quad C_V = T\left[\frac{\partial S}{\partial T}\right]_{V,N}$$

Ananda
28/2/25

Mid Sem Test

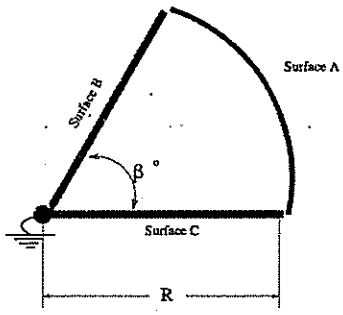
PH3202, Spring 2025 — DPS, IISER Kolkata

Instructor: Dr. Rajesh Kumble Nayak
10:00, 05 March, 2025.
Duration 90 Min.

- Read each question carefully.
- Answer all questions.
- Ensure all parts of the answer for a given question are in the same place.
- Write neatly and do all rough work on the answer sheet.
- Do not write on the question paper.
- Keep answers concise and to the point.

Good luck!

Question-1 [20 Marks]



Find the electrostatic potential in a chrage free region bound by the surfaces A, B and C as shown in the figure. Surface B and C are ground ($\Phi = 0$), Surface A is kept at potential $\Phi(\rho = R, \phi) = \sin\left(\frac{m_0 \pi \phi}{\beta}\right)$, where β is the angle between surface B and C, m_0 is a $+ve$ integer.

- Find the componets of electric field.
- Find the surface charge density close to the origin.

Question-2 [20 Marks]

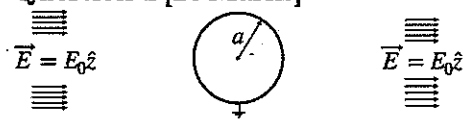
For the differential equation

$$y'' + \frac{2}{x}y' + y = S(x)$$

on the interval $\pi/2 < x < \pi$, the boundary conditions are $y(\pi/2) = 0$ and $y(\pi) = 0$

- Find the Green function $G(x; x_0)$ for the problem. Hint: homogeneous solutions are $x^{-1}\sin x$ and $x^{-1}\cos x$
- Use the Green function to find the solution in the case $S(x) = 1/x$.

Question-2 [20 Marks]



A conducting ground sphere of radius 'a' is placed in uniform electric field E_0, as shown in figure. Find the potential outside sphere.

The End

Midsem

PH 4202 Adv. Stat. Phys.

Duration 90 minutes

Max Marks 20

1. Explain, using the energy-entropy argument for the free energy, why there is no phase transition in the one-dimensional Ising model at finite temperature.

(2 marks)

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2. Consider the spin-1 Ising model in d dimensions with nearest neighbor coupling described by the Hamiltonian

$$H = -J \sum_{<i,j>} S_i S_j - h \sum_i^N S_i.$$

Here h is the applied external field. Given that the spins at each site S_i can take one of the three values $\{-1, 0, 1\}$ on the N lattice sites, derive the mean field equation for the order parameter. Take the number of nearest neighbours to be z.

(5 marks)

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3. Determine the critical exponent for the function $f(\varepsilon) = A\varepsilon^2 e^{-\varepsilon}$ as $\varepsilon \to 0$.

(1 mark)

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4. Consider the Landau free energy

$$L(\eta) = \frac{1}{2}a(T - T_c)\eta^2 + \frac{1}{4}b\eta^4 + \frac{1}{6}c\eta^6, \quad (\text{given } a, c, T, T_c > 0)$$

(a) What is the order of the transition (first or second) when $b > 0$?

(b) Determine the critical exponents β and γ when $b = 0$.
[Recall: $\eta \sim -\varepsilon^\beta$ and $\chi \sim \varepsilon^{-\gamma}$, where $\varepsilon = (T - T_c)/T_c$. To determine χ, you can add a term to $L(\eta)$ coupling to the field, if you wish.]

(c) What is the temperature for the equlibrium phase transition when $b < 0$?

(d) Qualitatively explain how the system can show hysteresis when $b < 0$. (You don't necessarily need to do the algebra. Just highlight the region of the hysteresis window and support your answer pictorially.)

(1+4+4+3 marks)

Wednesday, March 05, 2025
Instructor: Bhavtosh Bansal

Subject: Research Methodology (PH4203)
Mid-Semester Examination 2025

Answer all questions. Full marks: 20

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1. (a) What are the differences between positivism and scientific materialism? (3)
   (b) What are the necessary conditions for proposing a hypothesis? (3)

2. An outbreak of diarrhoea in the IISER Kolkata hostel was reported. It was found that students could be divided into the following categories according to what they consumed on the day before the outbreak.

Group	Food item consumed	Diarrhoea
1	Toast, Mutton rezala, Chow Mein	Occurred
2	Khichdi, Toast, Mutton rezala	Occurred
3	Toast, Chowmein, Khichdi	Not occurred
4	Toast, Mutton rezala, Chowmein, Khichdi	Occurred
5	Toast, Chow Mein	Not occurred
6	Khichdi, Toast	Not occurred

   Can you draw any conclusion from this data as to the cause behind the outbreak? If yes, mention the food item(s) responsible and explain your methodology for arriving to this conclusion in terms of John Stuart Mill's canons of induction. (3)

3. Are the following reasonings valid? Explain your answer briefly.

   (a) Premise 1: Infection by COVID virus leads to death within a month
   Premise 2: Ching Hu is not dead
   $\therefore$ Ching Hu has not been infected by COVID virus in the past month (1)

   (b) Premise 1: Infection by COVID virus leads to death within a month
   Premise 2: Ching Hu died today
   $\therefore$ Ching Hu was infected by COVID within the past month (1)

   (c) Premise 1: All insects have six legs
   Premise 2: Elephants are not insects
   $\therefore$ Elephants do not have six legs. (1)

4. Suppose you are performing an experiment to measure the resistivity of a wire made of a new substance that you have invented. The measured values (in MΩm) are: {25.3, 24.8, 26.1, 25.7, 24.9, 26.4, 20.1, 25.2, 26.0, 27.1, 25.5, 26.3, 25.8, 24.7, 26.9, 25.1, 26.7, 24.6, 27.2, 25.4, 26.5, 25.0, 26.6, 25.9, 27.0, 35.5, 24.5 }

   (a) Draw a box-and-whisker plot. Identify if there is any outlier. (2)
   (b) If you have to report measured value in a physics journal, how would you represent it with an error bar? (3)
   (c) With what confidence level can you state that the actual value is below 26.3 MΩm? (3)

Standard Normal Probabilities

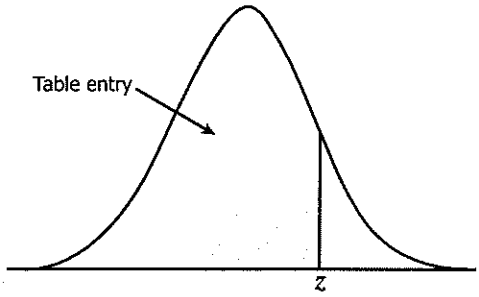

Table entry for z is the area under the standard normal curve to the left of z.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Table 1: The z-table for positive values of z

6/3

Date : 6th Mar, 2025 Mid-Sem

PH4204 : High Energy Physics

Duration: 90 minutes Total marks: 20

Time management tip: One marks is about 4 minutes (4.5 min to be exact).
First try the problem you are most comfortable with.
Useful relations are given at the back of the question paper.

1. **Quark Model:** Baryons are made three quarks, where quark is a spin-half fermion. We know that $\Delta^{++}(uuu)$ is made of three u quarks and it's $|\frac{3}{2}, +\frac{3}{2}\rangle$ spin state's wave-function is given by $|u\uparrow u\uparrow u\uparrow\rangle$. Based on symmetry arguments alone, write down the wave function of
   (a) Δ^+ in $|\frac{3}{2}, +\frac{1}{2}\rangle$ spin state. 2
   (b) Δ^0 in $|\frac{3}{2}, -\frac{1}{2}\rangle$ spin state. 2

2. **Alternate Proton:** Three quarks state's flavor part decomposes into 10-dim symmetric $F(S)$, 8-dim mixed-symmetric $F(M_S)$, 8-dim mixed anti-symmetric $F(M_A)$ and 1-dim anti-symmetric. While the spin part of the wavefunction decomposes into 4-dim symmetric $\chi(S)$, 2-dim mixed-symmetric $\chi(M_S)$ and 2-dim mixed anti-symmetric $\chi(M_A)$. Consider the *alternate* wave-function for spin-up proton to be

$$|p\uparrow\rangle = \frac{1}{\sqrt{2}}\left[F(M_S)\times\chi(M_A) - F(M_A)\times\chi(M_S)\right]$$

   (a) Write down the wave-function of above define proton in spin up state in terms of quarks and spins as done in the class. 4
   (b) Assuming the magnetic moments of quarks to be $\mu_u = -2\mu_d$ obtain the ratio of magnetic moments of proton and neutron. Results for neutron can be obtained by exchanging u and d quarks. 2

3. **QED scattering:** Consider massless electrons and massless muons interacting via massless photon.
   (a) Write down the Feynman rules for propagator and vertices 1
   (b) For the process $e^-(k_1)+\mu^-(p_1)\to e^-(k_2)+\mu^-(p_2)$, write down the matrix element and matrix element squared summed over spin. 1+2
   (c) Define $s = (p_1+k_1)^2$ (sum of intial state momentum), $t = (k_1-k_2)^2$ (difference of one of intial state and one final state momentum) and $u = (k_1-p_2)^2$ (difference of the initial state with other final state momentum). Rewrite the above computed expression for spin-summed matrix element square in terms s, t and u. 2
   Hint: Use momentum conservation to find other expressions for s, t and u and use them to get a compact expression.
   (d) For the process $e^-(k_1)+e^+(k_2)\to\mu^-(p_1)+\mu^+(p_2)$, write down the matrix element and matrix element squared summed over spin. 1+1
   (e) Define s, t and you in similar spirit as above, i.e. $s = (k_1+k_2)^2$ (sum of initial state momentum) etc. Rewrite the spin-summed matrix element squared for the annihilation process above. 1
   (f) Compare the expression for spin-summed expressions for above two processes in terms of their respective s, t, u variables and comment on any similarity. 1

Useful relations:

$F(M_S) = \frac{1}{\sqrt{6}}(udu + duu - 2uud)$ and $F(M_A) = \frac{1}{\sqrt{3}}(udu - duu)$

Similar for $\chi(M_S)$ and $\chi(M_A)$ can be obtained by replacing u/d with $\uparrow / \downarrow$

$\mathrm{Tr}\left[\gamma^\mu \gamma^\nu\right] = 4\eta^{\mu\nu}$
$\mathrm{Tr}\left[\gamma^\mu \gamma^\nu \gamma^\alpha \gamma^\beta\right] = 4\left(\eta^{\mu\nu}\eta^{\alpha\beta} - \eta^{\mu\alpha}\eta^{\nu\beta} + \eta^{\mu\beta}\eta^{\alpha\nu}\right)$

Indian Institute of Science Education and Research Kolkata

Department of Physical Sciences

PH4205 : General Theory of Relativity and Cosmology
Mid-semester Examination (Spring 2025)
Marks: 20, Duration: 90 minutes

1. Consider the coordinate transformation $\bar{x} \equiv (r, \psi) \rightarrow \bar{x}' \equiv (x, y)$ such that $x = \frac{1}{2} r \cos 2\psi$ and $y = \frac{1}{2} r \sin 2\psi$. Compute the elements of the transformation matrix Λ if $d\bar{x}' = \Lambda \, d\bar{x}$. If the invariant distance in $\bar{x}'$ coordinate system is $ds^2 = dx^2 + dy^2$ then compute the metric tensor in $\bar{x}$ coordinate system. [3+4]

2. For a metric tensor $g_{\mu\nu}$, if $\partial_\alpha g = g \, g^{\mu\nu} \, \partial_\alpha g_{\mu\nu}$ where g is the determinant of the metric then prove the following identity [4]

$$\nabla_\nu u^\nu = \frac{1}{\sqrt{-g}} \partial_\nu \left(\sqrt{-g} \, u^\nu \right) .$$

3. If $\nabla_\alpha g^{\mu\nu} = 0$ and $[\nabla_\alpha, \nabla_\beta] V^\mu = R^\mu{}_{\nu\alpha\beta} V^\nu$ then show that $R_{\mu\nu\alpha\beta} = -R_{\nu\mu\alpha\beta}$ where $g^{\mu\nu}$ is the inverse metric and V^μ is a contra-vector. [3]

4. By computing the Christoffel connections, find the geodesics on a space with the metric $ds^2 = dr^2 + r^2 d\theta^2$. What kind of curves are these geodesics? [2+2+2]

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PH 4207, Mid Sem Exam (Total Marks 20)
(24/02/25)

1. (a) Given the Bloch state on the Bloch sphere, $|\psi\rangle = cos\frac{\theta}{2}|0\rangle + exp(i\phi)sin\frac{\theta}{2}|1\rangle$, find the antipodal state ($|\tilde{\psi}\rangle$) of this Bloch state. Show explicitly using unitary rotation on the Bloch sphere, how can one go from the state $|1\rangle$ in the Bloch sphere to $|\tilde{\psi}\rangle$ in **exactly one** step. Write down explicitly the axis about which this rotation is to be performed. [4]

   (b) In the above example, please also show using Euler decomposition, how one can reach $|\tilde{\psi}\rangle$ from $|1\rangle$ on the Bloch sphere. You can use a maximum of up to 3 rotations/decomposition. [2]

2. Consider the composite system of two $spin\frac{1}{2}$ particles, represented by a suitably normalised state $|\psi\rangle = \alpha\,|00\rangle + \beta\,|01\rangle + \gamma\,|10\rangle + \delta\,|11\rangle$. If the measurement is performed on the first particle along the x- direction, discuss the effect of measurement on the wave function. What happens to the second particle in each case? [3]

3. (a) Determine the Schmidt decomposition of the two-qubit state,
   $|\phi_{AB}\rangle = \frac{1}{\sqrt{2}}\left[|0_A\rangle|0_B\rangle - \frac{3i}{5}|0_A\rangle|1_B\rangle - \frac{4i}{5}|1_A\rangle|1_B\rangle\right]$
   [4]

   (b) For another state ($|\psi_{AB}\rangle$) perform the singular value decomposition, and examine the entanglement of this state.
   $|\psi_{AB}\rangle = \frac{1}{\sqrt{2}}\left[|0_A\rangle|0_B\rangle - |0_A\rangle|1_B\rangle + |1_A\rangle|0_B\rangle + |1_A\rangle|1_B\rangle\right]$ [3]

4. Starting from the Bloch vector representation of density matrix for an electron, derive the equation of motion governing the Bloch vector of the electron, if it is subjected to an external magnetic field $\vec{B}$. Follow the **density matrix formulation**, covered in class.

   [4]

Evolutionary Dynamics (PH4209) Mid-Sem exam (Feb.27, 2025)

Answer all questions. Total Time=1.5 hours. Total Marks=20.The mark for each question is specified in brackets.

1. a) For a population consisting of two types of strategies A and B whose frequency dependent fitness's are f_A and f_B respectively, write down the replicator equation (assuming no mutation) showing the rate of change of frequency of type A. If a>c and d>b, obtain the possible equilibrium frequencies for A. a,b,c,d correspond to the matrix elements E(A,A), E(A,B), E(B,A), E(B,B) respectively of the payoff matrix. (2)

b) Prove the mixed state equilibrium solution obtained in a) is an *unstable* equilibrium. (2)

c) Consider a game between TFT, TF2T and ALLD where the expected number of rounds is $\overline{m}$. TF2T is a more cooperative strategy than TFT that cooperates with a defector twice before starting to defect.

(i) Write down the payoff matrix for such a game and derive the condition for which TFT is an ESS that can resist invasion by TF2T in a game between TFT, TF2T and ALLD strategies. (3)

d) In a TFT-ALLC-ALLD game, derive the expression for the critical frequency of ALLC players above which a small fraction of ALLD players can invade a mixed population of ALLC and TFT players. (2)

2. A finite population of size N is made up of two types of individuals A and B whose initial numbers in the population are **i** and **N-i** respectively. The population undergoes evolution by the Moran process.

a) If there is no difference between fitness between types A and B, what is the probability of choosing A for death and B for reproduction? (1)

b) If $\mathbf{x_i}$ is the fixation probability of type A, write down the equation of $\mathbf{x_i}$ in terms of $\mathbf{x_{i-1}}$ and $\mathbf{x_{i+1}}$ and solve the above equation to obtain the fixation probability of A starting from an initial state where just ***one*** A-type is present in the population along with N-1 other B-types. (3)

3. Consider two stochastic strategies with last round memory defined by S1(p1,p2,p3,p4) and S2(q1,q2,q3,q4) where p1,p2,p3,p4 are S1's probabilities of cooperating when the state of the game in the last round was CC, CD, DC, DD respectively.

(a) Write down the matrix giving the transition probabilities between the 4 possible Markov states. (2)

(b) If μ is the mutation probability of making a mistake (behavioural noise); obtain the transition probabilities in going from (i) *CD to DC* (ii) *DD to CC* in the presence of such behavioural noise. Justify each term in your expressions for the transition probabilities (4)

(c) In a game of WSLS vs ALLD, where $\overline{m}$ is the average number of rounds, obtain the condition on the parameters of the payoff matrix for WSLS to be stable against invasion by ALLD. (1)

PH4213 Mid Semester Examination

Total Marks : 20 **Time : 90 minutes**

Attempt all questions

Q 1) a) The centralizer of $H < G$, denoted by $C_G(H)$ is defined by

$$C_G(H) = \{g \in G : gh = hg \, \forall h \in H\}$$

Check whether

~~Show that~~ the centralizer is a normal subgroup of ~~$N_G(H)$.~~ G.

b) If A and$B$ are normal subgroups of G with only one element in common, show that every element of A commutes with every element of B.

c) Find the center of $SU(2)$ - the set of all elements that commute with every element of $SU(2)$. $[4+2+4]$

Q 2) The character table for the group T_d is given below (the name of the group and the symbols for the representations and the classes are not relevant for this problem - except as a mode of reference)

	E	$8C_3$	$3C_2$	$6\sigma_d$	$6S_4$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
E	2	-1	2	0	0
T_1	3	0	-1	-1	1
T_2	3	0	-1	1	-1

a) What is the order of this group?

b) Explicitly verify that each of the representations whose characters are listed in the table are irreducible.

c) Determine the number of times each irrep occurs in the reduction of a representation with characters $(17, -1, 5, 1, 3)$.

d) Determine the number of times each irrep occurs in the reduction of the direct product representation $T_1 \otimes T_2$.

The notation $8C_3$, for example, means that there are 8 elements in the class. $[1+2+3+4]$